

1. Multiple choice questions.

(i) Sort each circumstance as either “suffering from water scarcity” or “not suffering from water scarcity” based on the facts provided below.

- (a) A region with high annual rainfall.**
- (b) A region having high annual rainfall and a large population.**
- (c) A region having high annual rainfall but water is highly polluted.**
- (d) A region having low rainfall and low population.**

Answer:

- (a) Not suffering from water scarcity**
- (b) Suffering from water scarcity**
- (c) Suffering from water scarcity**
- (d) Not suffering from water scarcity.**

(ii) Which one of the following statements is not an argument in favour of multipurpose river projects?

- (a) Multi-purpose projects bring water to those areas which suffer from water scarcity.**
- (b) Multi-purpose projects by regulating water flow help to control floods.**
- (c) Multi-purpose projects lead to large scale displacements and loss of livelihood.**
- (d) Multi-purpose projects generate electricity for our industries and our homes.**

Answer: (c) Multi-purpose projects lead to large scale displacements and loss of livelihood

(iii) Here are some false statements. Identify the mistakes and rewrite them correctly.

- (a) Multiplying urban centres with large and dense populations and urban lifestyles have helped in proper utilisation of water resources.**
- (b) Regulating and damming of rivers does not affect the river’s natural flow and its sediment flow.**

(c) In Gujarat, the Sabarmati basin farmers were not agitated when higher priority was given to water supply in urban areas, particularly during droughts.

(d) Today in Rajasthan, the practice of rooftop rainwater water harvesting has gained popularity despite high water availability due to the Indira Gandhi Canal.

Answer:

(a) Multiplying urban centres with large and dense populations and urban lifestyles have resulted in improper utilisation of water resources.

(b) Regulating and damming of rivers does affect the river's natural flow and its sediment flow.

(c) In Gujarat, the Sabarmati basin farmers were agitated when higher priority was given to water supply in urban areas, particularly during droughts.

(d) Today in Rajasthan, the practice of rooftop rainwater water harvesting popularity has declined due to high water availability from Indira Gandhi Canal.

2. Answer the following questions in about 30 words.

(i) Explain how water becomes a renewable resource.

Answer: Water is often regarded as a renewable resource because it continuously replenishes through the hydrological cycle. This cycle involves three key processes:

- Evaporation
- Condensation
- Precipitation

(ii) What is water scarcity and what are its main causes?

Answer: Many cities illustrate this issue. Water scarcity often results from a large and growing population, which increases the demand for water and leads to unequal access. A larger population needs more water not just for daily use but also to produce sufficient food. As a result, water resources are being overused to expand irrigation for dry-season farming. Irrigated agriculture consumes the most water, and many farmers rely on their own

wells and tube wells to boost productivity. This overuse of water resources has negatively impacted both water availability and food security.

3. Answer the following questions in about 120 words.

(i) Discuss how rainwater harvesting in semi-arid regions of Rajasthan is carried out.

Answer: In the semi-arid and arid regions of Rajasthan, such as Bikaner, Phalodi, and Barmer, traditional houses often featured underground tanks, known as tankas, for storing drinking water. These tanks, some as large as a room, could measure up to 6.1 metres deep, 4.27 metres long, and 2.44 metres wide. Integrated into a sophisticated rooftop rainwater harvesting system, tankas were typically built within the main house or courtyard. They were connected to sloping roofs via pipes, allowing rainwater to flow from the roofs into the tanks. The initial rain was usually not collected, as it helped clean the roofs and pipes, while subsequent rainwater was stored. This harvested rainwater could be preserved in the tankas until the next rainfall, providing a reliable source of drinking water, especially during dry periods and summer months. In this region, rainwater, or 'palar pani,' is valued as the purest form of natural water.